

ALL PAIRS SHORTEST PATHS

Assumption: No negative cycles

$(i,j) \rightarrow$ For each vertex, apply Dijkstra's

$$O(n)$$

since applying for every vertex

$$O(|V| \cdot |E| \log |V|)$$

complexity of Dijkstra's

Each time you extract min. vertex, update shortest distance to neighbors ("relaxation")
Each edge is relaxed once.
Updating min. heap $\rightarrow O(\log |V|)$ $\therefore$ finally $O(|E| \times \log |V|)$

$$O(n^3 \log n) \text{ in the worst case.}$$

$|E| \approx |V|^2$ for complete graph.

Q. Can we do better?

For the sake of simplicity, let us consider DAGs.

Shortest $s-t$ path = $\min_{j \in [1, K]} \{ \text{shortest } (s-u_j) \text{ path} + w_{u_j-t} \}$

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CLAIM : A shortest path between any pair of vertices has length at most $|V|$.

$$d(i,j) = \min_{u \in V} \{ d(i,u) + d(u,j) \}$$

$\min_{u \in V} \{ d(i,u) + d(u,j) \}$
 $\hookrightarrow$ could contain j
 $\therefore d(i,j)$ would contain a cycle

- $\rightarrow$ Fine if G is DAG
- $\rightarrow$ But if G is not a DAG, this could lead to cyclic dependence.

$$\hat{d}(i,j,V) : \text{shortest distance between } i \text{ and } j \text{ using all } V \text{ vertices.}$$

$$\uparrow$$

$V = \{1, \dots, |V|\}$

$\hat{d}(i,j,K) : \text{shortest distance between } i \text{ and } j \text{ using vertices } \{1, \dots, K\}$

$$\hat{d}(i,j,K) = \min \left\{ \begin{array}{l} \hat{d}(i,j,K-1) \\ \hat{d}(i,K,K-1) + \hat{d}(K,j,K-1) \end{array} \right\}$$

shortest path from i to j over $\{1, \dots, K\}$

shortest path from i to K using $\{1, \dots, K-1\}$

shortest path from K to j using $\{1, \dots, K-1\}$

eliminating K to prevent cyclic dependencies

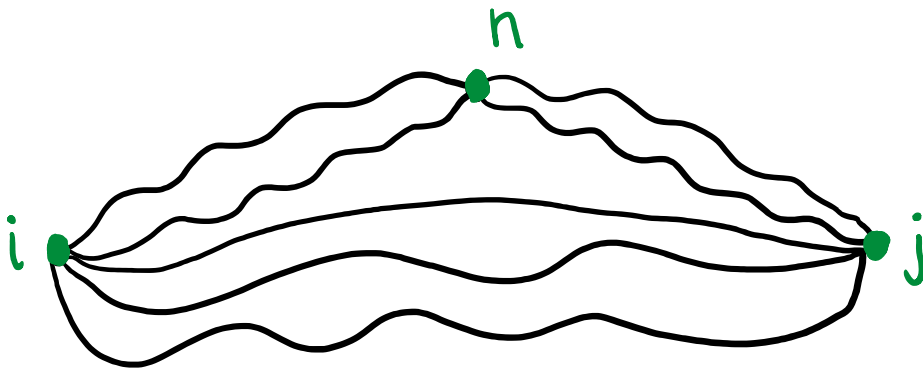
shortest path from i to j using $\{1, \dots, K-1\}$

Paths from i to j over $\{1, \dots, K\}$

$\rightarrow$

Paths that use vertex n

Paths that do not use vertex n



$$\hat{d}(i,j,K) = \min \left\{ \begin{array}{l} \hat{d}(i,j,K-1) \\ \text{shortest distance among all paths that go through } K \text{ and use } \{1, \dots, K\} \end{array} \right\}$$

$$\hat{d}(i,K,K-1) + \hat{d}(K,j,K-1)$$

= shortest distance among all paths from i to K using $\{1, \dots, K-1\}$ + shortest distance among all paths from K to j over $\{1, \dots, K-1\}$.

of all possible subproblems is $O(|V|^3)$

$\hat{d}(i,j,K)$

* Init: $\hat{d}(i,j,K) = \infty \quad \forall i,j,K$
 $\hat{d}(i,j,K) = w_{ij}$ if $(i,j) \in E$!!!

For k in $[1, |V|]$:
 For i in V :
 For $j \neq i$ in V :
 $\tilde{d}(i,j,k) = \min \left\{ \begin{array}{l} \hat{d}(i,j,k-1), \\ \hat{d}(i,k,k-1) + \hat{d}(k,j,k-1) \end{array} \right\}$
 Update $\hat{d}(i,j,k)$ if $\tilde{d}(i,j,k)$ is smaller.

When can $\hat{d}(i,j,1)$ make sense?
- If 1 is a neighbor of both i and j
- Or i or $j = 1$ and they are neighbors

$$\hat{d}(i,j,2) = \min \left\{ \begin{array}{l} w_{ij} \\ \hat{d}(i,j,1) \\ \hat{d}(i,2,1) + \hat{d}(2,j,1) \end{array} \right\}$$

Each entry of this table can be filled with $O(1)$ many lookups of already filled values.